

SIMULATION OF SEPARATING SHAPES ROBOTIC ARM

Khundet Beeche
School of Information Technology
Mae Fah Luang University
Chiang Rai, Thailand

Nontagorn Boayairuksa
School of Information Technology
Mae Fah Luang University
Chiang Rai, Thailand

Varisa Foo-trakul
School of Information Technology
Mae Fah Luang University
Chiang Rai, Thailand

Abstract— This project is a part of the Computer Engineering Pre-Project course. Separating Shapes Using A Robotic Arm. It is a popular technology in industrial and manufacturing applications because it can reduce costs, reduce errors, increase work efficiency, and increase safety. Especially in industries that require complex or repetitive work. For example, The use of robots in industrial production processes. This research presents an analysis and evaluation of the robotic arm in combination with the object recognition camera used to detect and classify various objects. It uses image processing technology to analyze the images received by the camera. Through the classification of various objects, the system will adjust for maximum accuracy and carry out the specified commands to be used to pick up objects precisely. Once all the joints and motors are connected, the robot arm is controlled by a microcontroller through the open cv program.

Keywords— robotic arm, classification, shape, computer vision

I. INTRODUCTION

Nowadays, automation technology is used in industry to increase the quality of production and perform tasks that humans cannot do. One such technology is industrial robots. Industrial robots imitate the human body. It is an imitation of specific parts of the body that will be used in industry. that is a human arm. Therefore, industrial robots are called by the word “robotic arms”, which means industrial robots that work like human arms and classified as a robot in the Handling operations/Machine tending group, which is the group with the highest usage [1].

At present, the development of technology allows robots to work in many ways. To respond to the industrial sector that needs more production capacity. For Thailand, it is important to Industry 4.0 that focuses on bringing digital technology and innovation into all aspects of society and the economy in Thailand. In order to enhance efficiency and sustainable economic growth for a longer period of time, the highlight is being able to connect individual consumer needs directly to the product production process [2].

To use the robotic arm, we must write a program to command the robotic arm to move accordingly. But simply using the robotic arm can mean less efficiency and versatility. Therefore, to increase the efficiency of the robotic arm We can use other technologies as well. So, in this study, we used a sensor, in this case, a camera, to capture an image of an object and classify it, then let the robot arm perform the programmed task. For example, if a rectangular object is detected, the robotic arm will pick it up and place it in the designated spot.

II. METHODOLOGY

A. Design

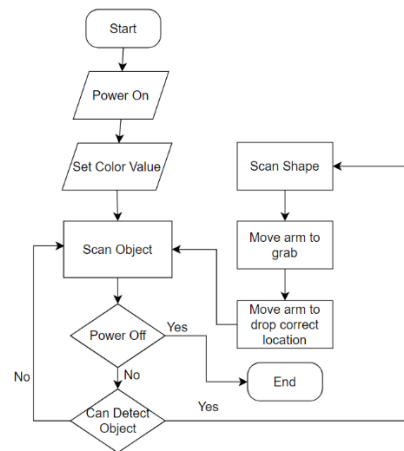


Fig. 1. Flowchart working steps of the robot arm system

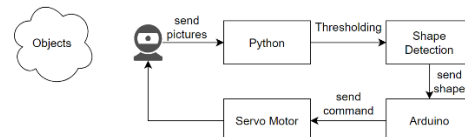


Fig. 2. Diagram working steps of the robot arm system

In part of the design of the program, it is divided into 2 main parts : the part of the program used to check the shape of the object and the part of the robotic arm that must work properly. For shape validation, we will use Python which is easy for beginners. coupled with OpenCV that will allow us to capture the color of objects We can adjust the color we want through the Trackbar. To make the color we chose white and other colors black as shown in Fig 2. We will hold the angle of the object to tell what shape the object is and the robotics arm will grab the object on the conveyor belt to sort it out.

The operation of the program is divided into the part of receiving the image of the camera for analyzing the image and the part of the robotic arm that receives the signal from the program. A soft real-time system that will work both when the working conditions are met. Once it is identified as a shape, it will command the robotic arm to move the object to the specified location as shown in Fig 2.

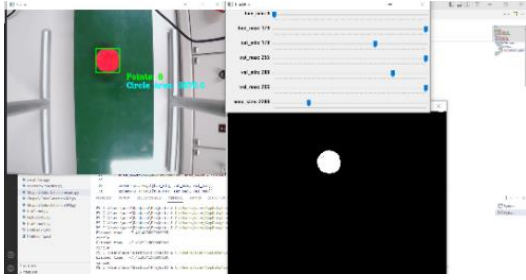


Fig. 3. Capture the color

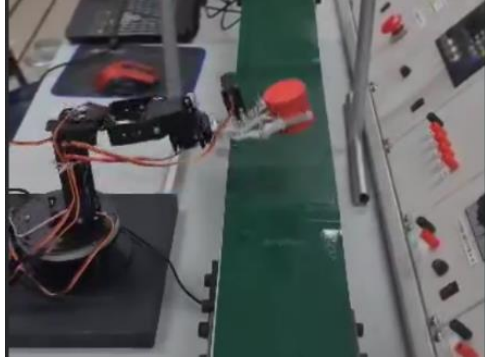


Fig. 4. Robotic arm's grasp of objects

B. Servo motor

The servo is responsible for controlling the operation of the machine. It can be used in a variety of ways, such as speed control. Control distance, etc. In use of a mechanical arm, it functions to control the direction of the mechanical arm and the holding of the mechanical arm, which has the following control principles.

In this project we are using a 6 DOF (Degrees of Freedom) robot arm consisting of 6 servo motors as shown in the picture. From the Fig 5, s1 is the base servo, s2 is the shoulder servo, s3 is the elbow servo, s4 is the wristed pitch servo, s5 is the wristed roll servo, and s6 is the gripper servo. Each servo is connected to the digital pin of the microcontroller. The microcontroller will control the servo motors according to the function that is defined.

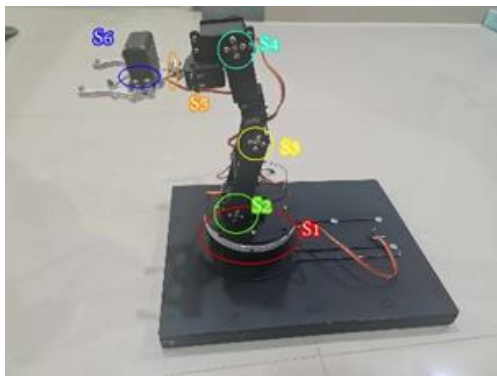


Fig. 5. Servo motor

C. Microcontroller

In this project, we must control 6 servo motors. We need a microcontroller that has ample I/O pins and flexibility with many libraries therefore we used Arduino mega 2560 R3. The advantage is high processing power, compatibility with most shields and libraries, and great for prototyping. The function

of Arduino mega 2560 R3 is to get values from the computer and calculate them for the control servo motor.

D. RGB color model

The RGB color model is an additive color model [3] in which the red, green and blue primary colors of light are added together in various ways to reproduce a broad array of colors. The name of the model comes from the initials of the three additive primary colors of red, green, and blue [4]. The main purpose of the RGB color model is for the sensing, representation, and display of images in electronic systems, such as televisions and computers.

If we combine it in various proportions it will obtain any color in the visible spectrum. in the present it likely to refers to a system that representing the colors used on a digital display screen.

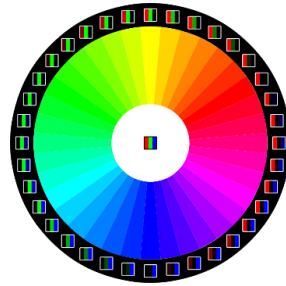


Fig. 6. RGB color wheel

E. HSL and HSV

HSL (hue, saturation, lightness) and HSV (hue, saturation, value) are alternative representations of the RGB color model. In both color system mixtures of the additive primary and secondary colors red, yellow, green, cyan, blue and magenta.

1) HSL

represents a way to simulate the mixing of different colors to create colors in the real world, with a brightness dimension analogous to varying amounts of black or white in the mix. Fully saturated colors are placed around a circle with a brightness value of 1/2, where a brightness value of 0 or 1 corresponds to completely black or completely white.

2) HSV

represents a simulation of how colors appear under light.

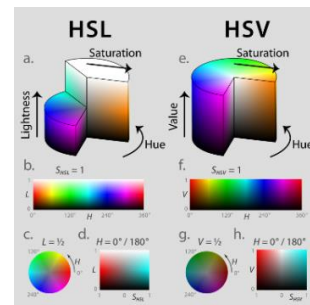


Fig. 7. HSL and HSV Cylinder

The difference between HSL and HSV is that the color with maximum brightness in HSL is pure white but the color with maximum brightness/brightness in HSV is similar to

shining white light on a colored object. Shining a dark light on a red object will still make the object appear red, just brighter and more intense, while shining a dark light on a red object will make the object appear darker and less bright.

F. Thresholding

Thresholding is the use of grayscale images to compare the pixels of the image in each area. It adjusts to 0 when it is less than a constant and to 255 when it is greater than a constant for color separation purposes.

Thresholding is a useful technique for improving image quality. Or used to detect certain objects or lines in an image, such as in detecting objects where the image is black and white. To clearly distinguish the object from the background. or in the distribution of edges of objects contained in the image.



Fig. 8. Thresholding

III. EXPERIMENT

From the purpose and scope of the research, The results of the testing of the robot arm's grasping accuracy will be discussed. In this test, values can be collected directly from the robot arm through a program in which an image of the object will show. When testing the accuracy of the robot arm, it will test the size of the object, the working time of the mechanical arm, and the distance between the robot arm and the webcam.

Data collection testing accuracy in picking workpieces is done using 3 objects circles, triangles, and rectangles. And will test picking up objects 20 times each. This data collection is a test of the robot arm's tolerances, the size of the object can grip, and the working time of the robot arm.

A. Size of the object

The size of the object affects the grasp of the robotic arm. Which must be of the appropriate size for holding by dividing the size of the object into 3 sizes with equal areas.

TABLE I. THE SIZE OF THE OBJECT THAT THE ROBOT ARM CAN HANDLE

Area / Object	Triangle	Rectangle	Circle
Less than 9 cm ²	✓	✗	✗
Between	✓	✓	✓
More than 12.25 cm ²	✗	✓	✓

B. Working time of the robot arm

Based on the objective and scope of the research, testing of the robot arm's grasping accuracy will begin to evaluate its efficiency and effectiveness. This test aims to measure and test important factors that affect the performance of the robot arm,

such as working accuracy, working speed, and object sorting efficiency.

a) *Conveyor speed* : The distance between the webcam and the robot arm is 25 cm from the point of the camera and the point of the robot arm. The mechanical arm needs time to grip as show in Table II.

TABLE II. THE AMOUNT OF TIME FOR ROBOTS ARM THAT MOVE IN THE SAME DIRECTION AT DIFFERENT SPEEDS

Conveyor speed	The amount of time the robotic arm uses
100%	4.81 s
75%	6.02 s
50%	7.23 s
25%	8.41 s

b) *Distance between the robot arm and webcam* : Time required by the robot arm at 100% Conveyor speed. The mechanical arm needs time to grip as follows.

TABLE III. TIME FOR GRASP OBJECTS WHEN THE BELT CONVEYOR CONTROLLED SPEED 100%

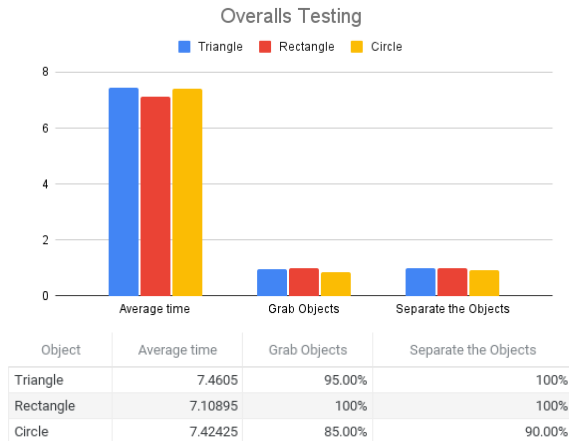
Distance	The amount of time the robotic arm uses
25 cm	4.81 s
30 cm	5.82 s
35 cm	6.71 s

c) Overalls Testing

In this experiment, data will be collected to test the accuracy of the robot arm, divided into a test to detect the accuracy of grasp objects and a test of the ability to separate the objects. The test data is collected using principles from computer vision to workpiece detected by the camera, after which the robot arm moves to find the object.

The results of this experiment provide information on the performance and evaluate the performance of the robot arm. The test results are shown in Table IV.

TABLE IV. OVERALLS TESTING



The robotic arm experiment used to sort objects uses robotic arm technology to sort various objects. It can be concluded that the robotic arm can separate various objects by using an image processing program to check the location and characteristics of the objects. Including controlling the mechanical arm with a program, allowing the operation of the mechanical arm to be adjusted appropriately, such as object size and duration.

IV. RESULT

In this experiment, the robot arm was tested for detecting and sorting objects in the arm's working time, object sorting accuracy and object grasping success. The results of this experiment provide information on the robot arm's performance and evaluate its performance and improve the robot system for working with objects that change with the environment.

The results of this experiment demonstrate the ability of the robot arm to perform successfully even though it is not 100% perfect in terms of effective working time and object sorting accuracy. This experiment will be useful in the development and improvement of robot technology in the future, allowing for greater application of robots in tasks that require more accurate and efficient gripping and sorting of objects.

V. DISCUSSION

Result of working on senior projects. Using cameras to detect and sort objects with complex shapes and appearances may cause the robot arm to not be able to sort objects well enough. This requires high image processing capabilities and may require mathematical methods to plan more efficient object capture and sorting. Including the detection of low-light objects may cause shadows on the objects. This makes object detection and sorting difficult because shadows can make objects look more similar. This is the only problem with using the camera in low-light environments, which is different from well-lit locations.

After testing the robotic arm The research team found problems with real-time Object Detection and Tracking for separating objects. We therefore solve the problem by adding a system to identify objects by the Area size function that will set the minimum size of the object so that the program can examine only the objects we want and use it to further develop a robot arm that can pick up a wider variety of objects. It can also be further developed in terms of control and faster frequency of grab objects.

Therefore, in this section, the research team has to improve and develop the next version of the system development.

VI. CONCLUSION

This project was created to use a robot arm that uses a camera to detect and determine the location of objects through image processing and can determine what the object looks like. Using a camera to detect and sort objects with the OpenCV program, which can lead to Continue to develop and improve the robot arm to work more efficiently. By improving the structure of the robot arm to allow a lower distance to release objects, change the motor to a higher output power and used to further develop the arm mounted on a mobile robot. Has the ability to handle a variety of objects with minimal deviation. It can also be applied to use robot

arms in other areas such as agricultural work. Industrial production work, etc.

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